

# Horn telescope stand assembly

Construction guide | 2021 design

Build a wooden stand for the horn telescope cradle. Notched base pieces allow the stand to come apart for transport.

## Materials

| Quantity  | Part                                                 |
|-----------|------------------------------------------------------|
| 4         | 2 × 4 boards, 3 feet long                            |
| 2         | 2 × 4 boards, 2 feet long                            |
| 2         | 2 × 4 braces, 2 feet long, with a 45° cut at one end |
| As needed | 2 1/2-inch construction screws                       |
| 2 each    | 3 1/2-inch-long 1/4-20 bolts, wing nuts, and washers |

### 1 Cut the base notches

Cut matching saddle notches in the base boards, as shown below. Each notch is 1 1/2 inches wide and 1 3/4 inches deep. Fit the notches together to form the base.



**Figure 1.** The matching notches let the base boards fit together.

## Assemble the stand

### 2 Attach the uprights

Use 2 1/2-inch screws to join the four 3-foot boards in two right-angle pairs. Position each joint 10 1/2 inches from one end, as shown in the finished stand.

### 3 Add the braces

Attach the two 2-foot braces to the 3-foot boards. Use the 45° ends to support the right-angle joints, as shown in Figure 2.

### 4 Attach the cradle

Connect the cradle to the stand with the 1/4-20 bolts, washers, and wing nuts. The completed telescope is shown in Figure 3.



Figure 2. Braced frame (left). Figure 3. Stand with the horn telescope attached (right).

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